

E-Commerce Backend API Using FastAPI

Technical Documentation

Intern details

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No. Of Weeks:- 4 weeks

Introduction

The E-Commerce Backend API is a web-based backend application developed using FastAPI and SQLite. The project provides RESTful API endpoints for managing users and products in an e-commerce environment. It demonstrates backend development concepts such as API creation, database integration, request validation, and CRUD operations.

Project Objective

The main objective of this project is to develop a scalable backend system for an e-commerce platform using FastAPI. The application allows users to register, add products, and retrieve product information through API endpoints.

Technologies Used

- Python
- FastAPI
- SQLAlchemy
- SQLite
- Pydantic
- Uvicorn
- Visual Studio Code
- GitHub

Project Features

- User Registration
- Product Creation
- Product Retrieval
- SQLite Database Integration
- RESTful API Architecture
- Interactive Swagger Documentation
- Data Validation using Pydantic

Project Structure

API Endpoints

User APIs

- - Register a new user

Product APIs

- - Add a new product
- - Retrieve all products

Home API

- - API Status Check

Working of the Project

1. The user sends requests through Swagger UI
2. FastAPI validates incoming data using Pydantic models
3. SQLAlchemy handles database operations
4. Data is stored in SQLite database
5. API responses are returned in JSON format
6. Swagger UI displays request and response details

Output

The application successfully performs:

- User Registration
- Product Creation
- Product Retrieval

API testing can be performed through: <http://127.0.0.1:8000/docs>

Advantages

- Fast and lightweight framework
- Easy API testing
- Simple database integration
- Beginner-friendly architecture
- Scalable backend design

Future Enhancements

- JWT Authentication
- Shopping Cart Module
- Order Management
- Payment Integration
- Admin Dashboard
- Product Search and Filtering

Conclusion

The E-Commerce Backend API project successfully demonstrates backend development using FastAPI. It provides a strong foundation for building modern e-commerce applications and helps in understanding API development, database management, and RESTful architecture.